

Planar Graphs Are 6-Colorable

1 Preliminaries

All graphs are finite and simple (no loops, no parallel edges). For a graph G we write $V(G)$ and $E(G)$ for its vertex and edge sets and $\deg(x)$ for the degree of a vertex x .

Definition 1. A graph is *planar* if it can be drawn in the plane such that edges are simple curves between their endpoints that meet only at common endpoints. Such a drawing is a *plane graph*. The *faces* of a plane graph are the connected components of the plane minus the drawing.

Definition 2. A k -*coloring* of a graph G is a map $c: V(G) \rightarrow \{1, \dots, k\}$ such that $c(x) \neq c(y)$ for every edge $xy \in E(G)$.

Every subgraph of a planar graph is planar: simply delete the corresponding points and curves from the drawing.

2 Euler's Formula

Lemma 3 (Euler's formula). *Let G be a connected plane graph with v vertices, e edges and f faces. Then $v - e + f = 2$.*

Proof. We proceed by induction on e . Since G is connected, $e \geq v - 1$.

If $e = v - 1$, then G is a tree. A drawing of a tree does not separate the plane, so $f = 1$ and $v - e + f = v - (v - 1) + 1 = 2$.

If $e \geq v$, then G is not a tree and hence contains a cycle C . Let uw be an edge of C . By the Jordan curve theorem, the drawing of C separates the plane into an inside and an outside region, and the two sides of the curve representing uw lie in different faces of G . Deleting uw therefore merges these two faces into one while leaving all other faces unchanged. The graph $G - uw$ is still connected (since uw lies on a cycle), and it has v vertices, $e - 1$ edges and $f - 1$ faces. By induction, $v - (e - 1) + (f - 1) = 2$, that is, $v - e + f = 2$. $\square$

3 Edge Density

Lemma 4 (Edge density). *Every planar graph G with $v \geq 3$ vertices has at most $3v - 6$ edges.*

Proof. Adding edges only increases e , so we may assume G is edge-maximal planar; in particular G is connected, since otherwise an edge joining two components could be added to the drawing without crossings. Fix a plane drawing of G with e edges and f faces.

Consider for each face its boundary walk and count the number of edge-sides it contains. Every edge has two sides, each of which lies on exactly one face boundary, so the total count is exactly $2e$. On the other hand, since G is simple, connected and has $v \geq 3$ vertices (and thus $e \geq 2$), every face

boundary contains at least 3 edge-sides: a boundary with only one or two edge-sides would require a loop, a pair of parallel edges, or a graph consisting of a single edge. Hence $2e \geq 3f$, i.e. $f \leq \frac{2}{3}e$.

By Euler's formula (Lemma 3),

$$2 = v - e + f \leq v - e + \frac{2}{3}e = v - \frac{1}{3}e,$$

which rearranges to $e \leq 3v - 6$. □

4 A Vertex of Small Degree

Lemma 5 (Low-degree vertex). *Every nonempty planar graph G has a vertex of degree at most 5.*

Proof. Let $v = |V(G)|$ and $e = |E(G)|$. If $v \leq 6$, every vertex has degree at most $v - 1 \leq 5$. Otherwise $v \geq 3$, and by the edge density bound (Lemma 4),

$$\sum_{x \in V(G)} \deg(x) = 2e \leq 6v - 12 < 6v.$$

So the average degree is less than 6, and some vertex has degree at most 5. Equivalently: not all vertices can have high degree (≥ 6). □

5 The Six Color Theorem

Theorem 6 (Six color theorem). *Every planar graph admits a 6-coloring.*

Proof. By induction on the number of vertices v of the planar graph G . If $v = 0$, the empty map is a 6-coloring.

Let $v \geq 1$. By Lemma 5, G has a vertex x with $\deg(x) \leq 5$. The graph $G - x$ is a subgraph of G , hence planar, and has $v - 1$ vertices. By induction it has a 6-coloring c . The neighbors of x use at most 5 distinct colors under c , so some color $i \in \{1, \dots, 6\}$ is not used by any neighbor of x . Extending c by $c(x) = i$ yields a 6-coloring of G . □